

Chapter 6: Number Play

Worksheet 6: Number Play

Learning Objective: Apply divisibility rules, identify prime and composite numbers, find prime factorisation, and use HCF and LCM to solve real-life grouping and scheduling problems.

Worked Example:

Find the prime factorisation of 84.

Solution: $84 = 2 \times 42 = 2 \times 2 \times 21 = 2 \times 2 \times 3 \times 7 = 2^2 \times 3 \times 7$.

Questions (1–20)

1. Is 97 a prime number?
2. List all factors of 36.
3. Is 84 divisible by 6? Use a divisibility rule to check.
4. What is the next number in this pattern: 1, 4, 9, 16, _____?
5. Find the prime factorisation of 60.
6. Find the prime factorisation of 96.

7. Find the HCF of 24 and 36 using prime factorisation.

8. Find the LCM of 12 and 18.

9. Check whether 144 is divisible by both 8 and 9.

10. List the prime numbers between 50 and 70.

11. Find the HCF and LCM of 18, 24 and 36.

12. A number is divisible by both 4 and 9. What is the smallest such number?

13. Find the sum of the first 5 prime numbers.

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14. A school arranges 90 students into rows of equal size, with each row having no fewer than 5 and no more than 15 students. List the possible row sizes.

 15. Find the prime factorisation of 1,000 and express it in exponent form.

 16. Two bells ring every 12 minutes and 18 minutes respectively, starting together at 9:00 am. After how many minutes will they ring together again, and at what time?

 17. A gardener has 48 rose plants and 60 marigold plants, to be planted in identical rows with the maximum possible plants of each type per row. Find the plants per row, and the number of rows of each type.

 18. Find a 4-digit number divisible by 3, 4 and 5 simultaneously, and explain your method.

 19. Describe a real-life situation where you would need the LCM (scheduling, packing in groups) or the HCF (sharing equally).

20. Check today's date (the day of the month). Is it a prime number, a composite number, or neither? Explain.

Reflection Question: Why do you think prime numbers are sometimes called the “building blocks” of all numbers? Write your own explanation.

Real-Life Application Question: Find a situation at home where things need to be grouped or shared equally (sweets, chairs, tiles). Explain how factors or HCF could help solve it.